



The Scots College

Student Number: _____

Teacher: _____

2023

Year 12 Assessment Task 2

Mathematics Extension 1

Date: Monday 1st May, 2023

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen only.
- Calculators approved by NESA may be used.
- A reference sheet is provided separately.
- Show all relevant mathematical reasonings and/or calculations.
- One A4 page of double-sided hand-written notes can be utilised.

Total marks: 52

- **Section I** (8 marks)
 - Allow about 10 minutes for this section.
 - Attempt all questions.
 - Answer on the multiple-choice answer sheet provided.
- **Section II** (44 marks)
 - Allow about 1 hour 20 minutes for this section.
 - Attempt all questions, providing mathematical reasoning/calculations where necessary.
 - Start each question in a **new** booklet.

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Section I

8 marks

Attempt Questions 1–8.

Allow about 10 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 8. Circle the correct response.

1. Which of the following expressions is equivalent to $2\cos 7x \sin 2x$?

(A) $\sin 9x - \sin 5x$

(B) $\cos 5x + \cos 9x$

(C) $\sin 5x + \sin 9x$

(D) $\cos 9x - \cos 5x$

2. What is the angle between the vectors $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 18 \end{pmatrix}$?

(A) $\theta = \cos^{-1} \left(\frac{52}{65} \right)$

(B) $\theta = \cos^{-1} \left(\frac{-52}{65} \right)$

(C) $\theta = \cos^{-1} \left(\frac{33}{65} \right)$

(D) $\theta = \cos^{-1} \left(\frac{-33}{65} \right)$

3. The radius of a sphere is increasing at a rate of 6 centimetres per minute. What is the rate of increase of the volume of the sphere, in cubic centimetres per minute, when the radius is 7 centimetres?

(A) 112π

(B) 144π

(C) 1168π

(D) 1176π

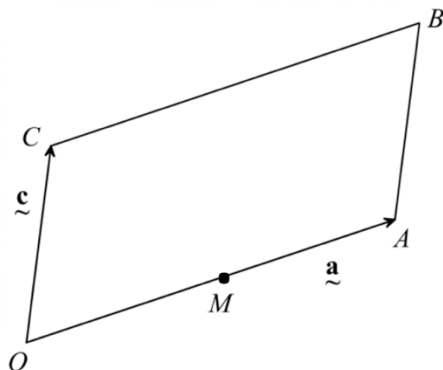
4. A body has displacement of $4\vec{i} + \vec{j}$ metres at a particular time. The body moves with constant velocity and three seconds later its displacement is $-5\vec{i} + 13\vec{j}$ metres.

Which of the following represents the velocity vector in ms^{-1} ?

- (A) $2\vec{i} + 6\vec{j}$
- (B) $-2\vec{i} + 2\vec{j}$
- (C) $4\vec{i} - 3\vec{j}$
- (D) $-3\vec{i} + 4\vec{j}$

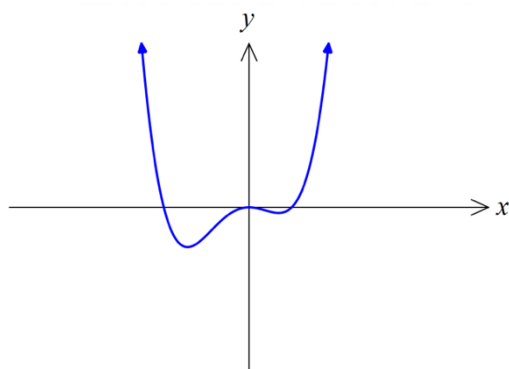
5. In the diagram, $OABC$ is a parallelogram. The vector $\vec{OA} = \vec{a}$ and $\vec{OC} = \vec{c}$.

M is the midpoint of $\vec{OA}$. Which of the following expressions is represented by the vector $\vec{MC}$?



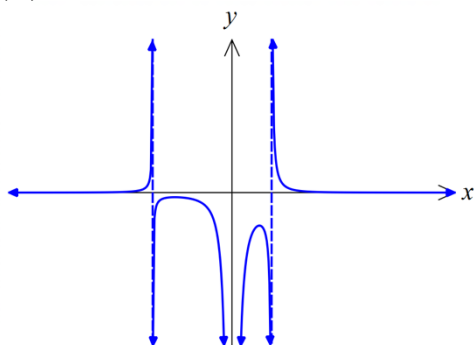
- (A) $\vec{a} + \frac{1}{2}\vec{c}$
- (B) $\vec{c} - \frac{1}{2}\vec{a}$
- (C) $\frac{1}{2}\vec{a} + \vec{c}$
- (D) $\frac{1}{2}\vec{a} - \vec{c}$

6. The diagram shows the graph of $f(x)$

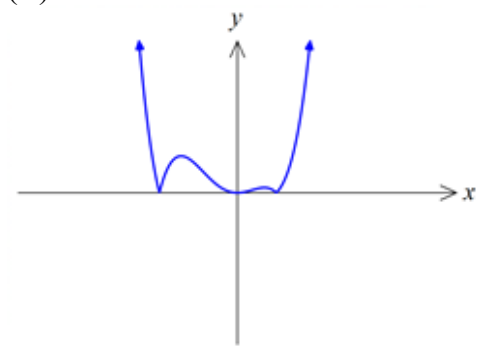


Which graph best represents $y = f(|x|)$?

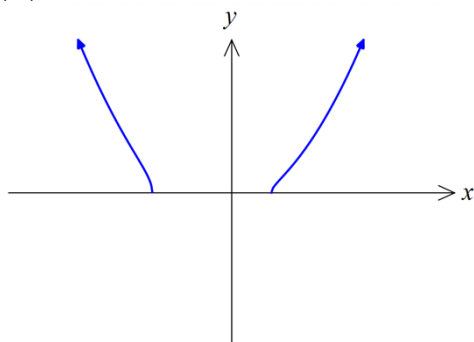
(A)



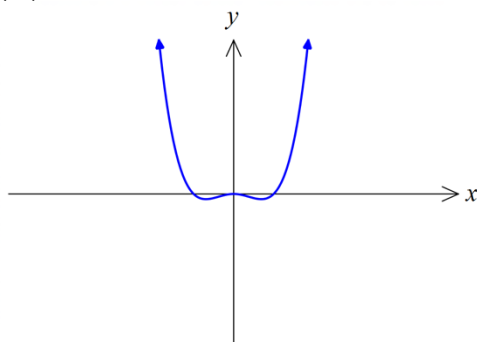
(B)



(C)



(D)



7. A and B are the points $(-4, 1)$ and $(5, 13)$ respectively. What is the unit vector in the direction of $\overrightarrow{AB}$?

(A) $4\hat{i} + 3\hat{j}$

(B) $\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$

(C) $9\hat{i} + 12\hat{j}$

(D) $\frac{2}{5}\hat{i} - \frac{4}{5}\hat{j}$

8. What is the coefficient of x^3 in the expansion of $(2 - 3x)^7$?

(A) -864

(B) 864

(C) $-15\,120$

(D) $15\,120$

Section II

44 marks

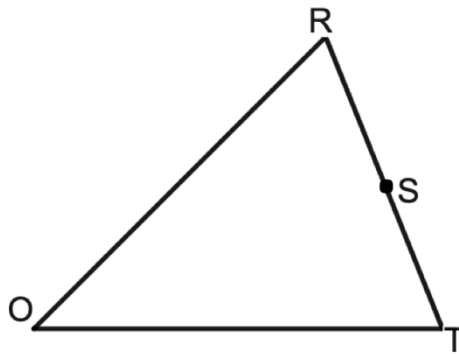
Attempt Questions 9, 10 and 11 **writing your answers in separate booklets for each question.**
Allow about 1 hour 20 minutes for this section.

Question 9 (14 marks) Start a new answer booklet.

Mark

(a) Use the substitution $t = \tan \frac{x}{2}$ to show that $\sec x + \tan x = \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)$ **3**

- (b) S is a point on the side RT of $\triangle ORT$. **3**
Given that OS is the perpendicular bisector of RT , prove using vectors that $\triangle ORT$ is isosceles.



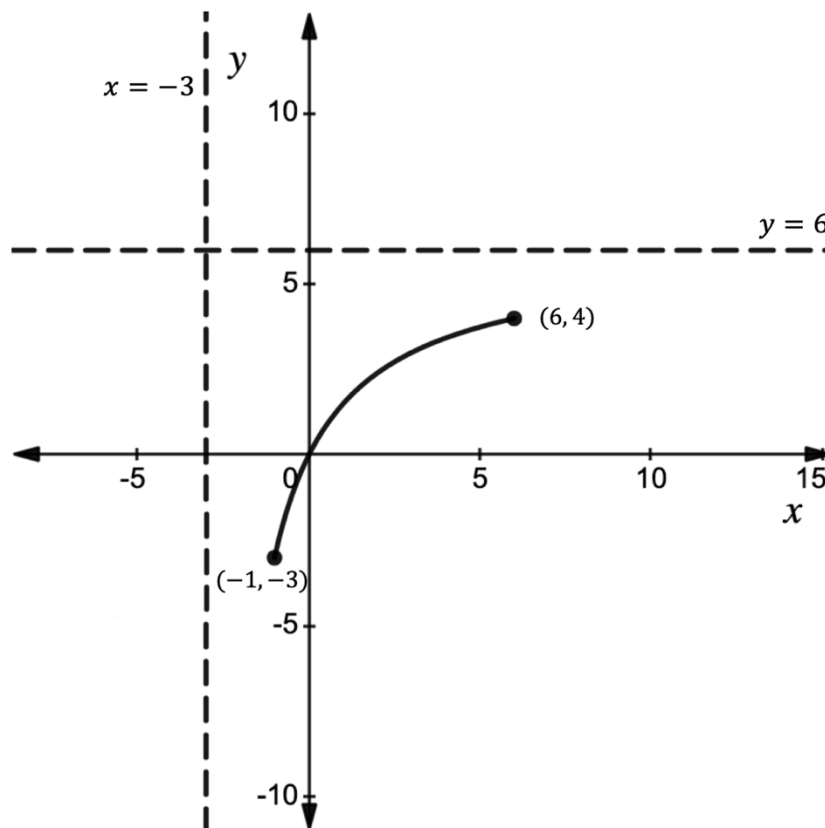
Not to Scale

(c) A car is driving with velocity of $(7\tilde{i} - 5\tilde{j})ms^{-1}$

- (i) Find the exact speed of the car. **1**
(ii) Find the bearing the car is travelling on to the nearest degree. **2**

Question 9 continues on page 8

- (d) The diagram below shows a sketch of the graph $y = f(x)$,
 where $f(x) = \frac{6x}{x+3}$ with domain $-1 \leq x \leq 6$

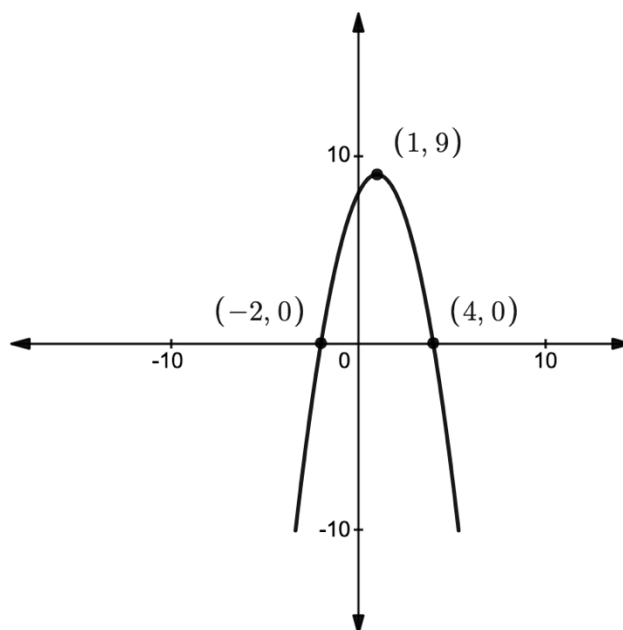


- | | | |
|-------|---|---|
| (i) | Copy or trace this diagram into your booklet.
On the same set of axes, sketch the graph of the inverse function, $y = f^{-1}(x)$. | 2 |
| (ii) | Find the coordinates of any points of intersection of the two curves $y = f(x)$ and $y = f^{-1}(x)$. | 1 |
| (iii) | Find an expression for $y = f^{-1}(x)$ in terms of x . | 2 |

- (a) The diagram shows the graph of $f(x) = -x^2 + 2x + 8$

3

In the answer booklet, sketch $y = \frac{1}{|f(x)|}$



- (b) Prove by mathematical induction that for all integers $n \geq 1$,

3

$9^{n+2} - 4^n$ is divisible by 5.

- (c) Given that $f(x) = 5 \sin x + 12 \cos x$

- (i) Express $5 \sin x + 12 \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$

3

- (ii) Find the maximum value of $5 \sin x + 12 \cos x$.

1

- (iii) Hence, find the smallest positive value of x , for which this maximum occurs.

1

- (d) If $\vec{a} = -3\vec{i} + \vec{j}$ and $\vec{b} = -m\vec{i} - \vec{j}$ are vectors, where $m \in \mathbb{R}$.

3

Find the value(s) of m if vector $\vec{b}$ is perpendicular to vector $\vec{a} - \vec{b}$

(a) A curve has the parametric equations

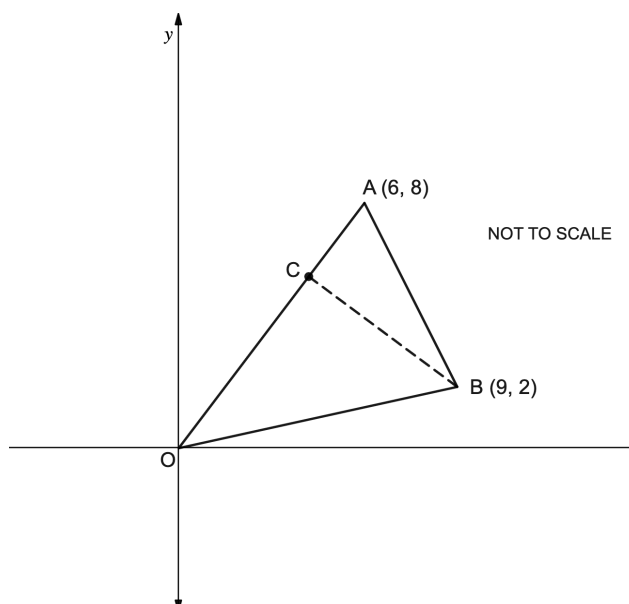
$$x = \sin^2 t, \quad y = \sin 2t, \quad 0 < t < \pi$$

(i) Show that $\frac{dy}{dx} = 2 \cot 2t$ 3

(ii) Find a cartesian equation for the curve 3

(b) The coefficient of x in the expansion of $\left(x + \frac{1}{ax^2}\right)^7$ is $\frac{3}{7}$. Find all possible values of a . 3

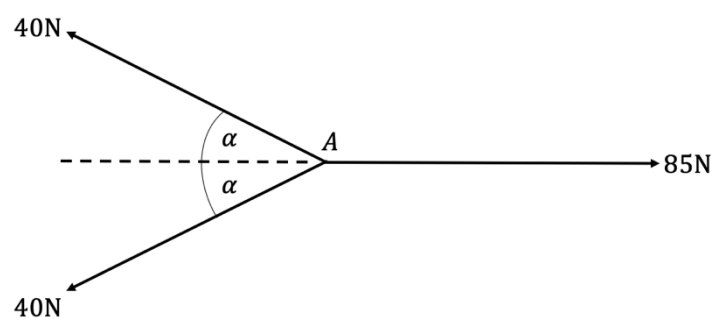
(c) The diagram below shows $\triangle OAC$, where $\vec{OA} = 6\vec{i} + 8\vec{j}$ and $\vec{OB} = 9\vec{i} + 2\vec{j}$.



(i) By considering the projection of $\vec{OB}$ onto $\vec{OA}$, find the coordinates of C 2

(ii) Hence, calculate the exact area of $\triangle OAB$ 2

- (d) Three forces of magnitude 85N, 40N and 40N act at a point A , as shown on the diagram below.
The angle α is such that $\cos \alpha = \frac{3}{4}$



Find the magnitude and direction of the resultant vector.

3

End of Examination 😊

Student Number: Solutions

Teacher: _____

Year 12 Mathematics Extension 1

Section I Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct (arrow pointing to B)

Question 1	A ☒	B ☐	C ☐	D ☐
Question 2	A ☒	B ☐	C ☐	D ☐
Question 3	A ☐	B ☐	C ☐	D ☒
Question 4	A ☐	B ☐	C ☐	D ☒
Question 5	A ☐	B ☒	C ☐	D ☐
Question 6	A ☐	B ☐	C ☐	D ☒
Question 7	A ☐	B ☒	C ☐	D ☐
Question 8	A ☐	B ☐	C ☒	D ☐

Question 9

a) LHS = $\sec x + \tan x$

$$= \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2} \quad (1)$$

$$= \frac{t^2 + 2t + 1}{1-t^2}$$

$$= \frac{(t+1)^2}{(1-t^2)(1+t)}$$

$$= \frac{t+1}{1-t} \quad (2)$$

RHS = $\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$

$$= \frac{\tan \frac{\pi}{4} + \tan \frac{x}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{x}{2}} \quad (1)$$

$$= \frac{1+t}{1-1 \times t}$$

$$= \frac{1+t}{1-t} \quad (2)$$

$$= \text{LHS}$$

b). Let $\vec{OR} = \underline{r}$, $\vec{OT} = \underline{t}$

$$\vec{RT} = \underline{t} - \underline{r}$$

$$\begin{aligned} \vec{OS} &= \underline{t} - \frac{1}{2} \vec{RT} \\ &= \underline{t} - \frac{1}{2} (\underline{t} - \underline{r}) \\ &= \frac{1}{2} (\underline{t} + \underline{r}) \end{aligned} \quad (1)$$

$$\vec{OS} \perp \vec{RT}$$

$$\begin{aligned} \vec{OS} \cdot \vec{RT} &= 0 \\ \frac{1}{2} (\underline{t} + \underline{r}) \cdot (\underline{t} - \underline{r}) &= 0 \\ \underline{t} \cdot \underline{t} - \underline{t} \cdot \underline{r} + \underline{r} \cdot \underline{t} - \underline{r} \cdot \underline{r} &= 0 \end{aligned} \quad (1)$$

$$\underline{t} \cdot \underline{t} - \underline{r} \cdot \underline{r} = 0$$

$$|\underline{t}|^2 - |\underline{r}|^2 = 0$$

$$|\underline{t}| = |\underline{r}| \quad (1)$$

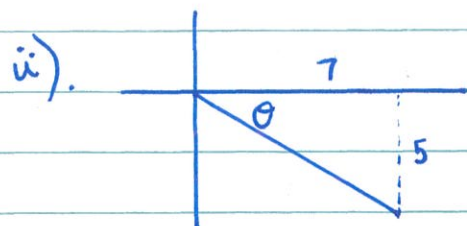
$\therefore \triangle ORT$ is isosceles since $OT = OR$.

c) $7\mathbf{i} - 5\mathbf{j}$

i) speed = $\sqrt{7^2 + (-5)^2}$

= $\sqrt{74}$

(1).



$\tan \theta = \frac{5}{7}$

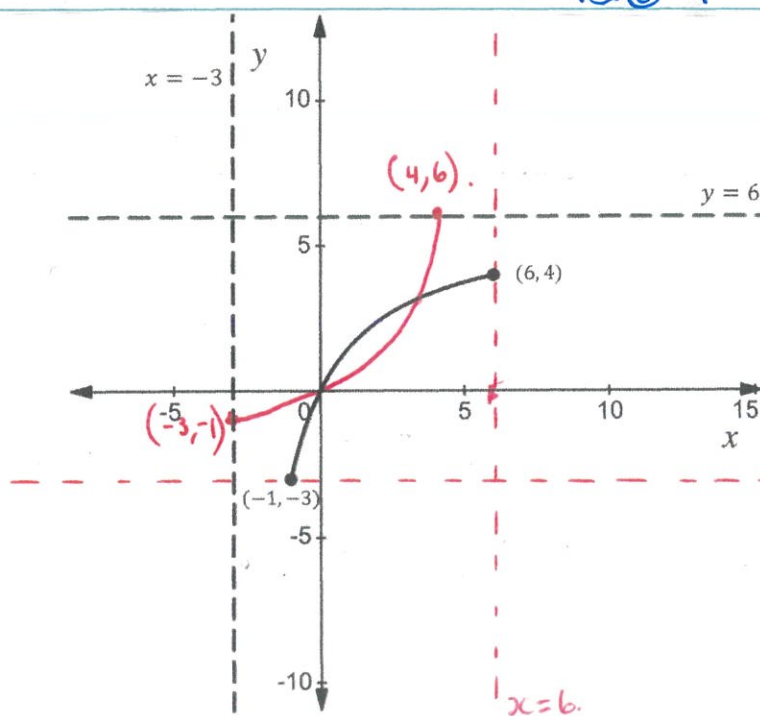
$\theta = 35.53767779^\circ$ (1)

= 36° (nearest degree).

bearing = $90 + 36$
= $126^\circ T$

(1).

d) i)



(1) asymptotes.

($\frac{1}{2}$) points

($\frac{1}{2}$) shape.

ii) $\frac{6x}{x+3} = x$

iii) $f^{-1}(x)$: $x = \frac{6y}{y+3}$

$6x = x^2 + 3x$

$x^2 - 3x = 0$

$x(x-3) = 0$

$x = 0, 3$

pt $(0,0)$, $(3,3)$. (1).

$xy + 3x = 6y$

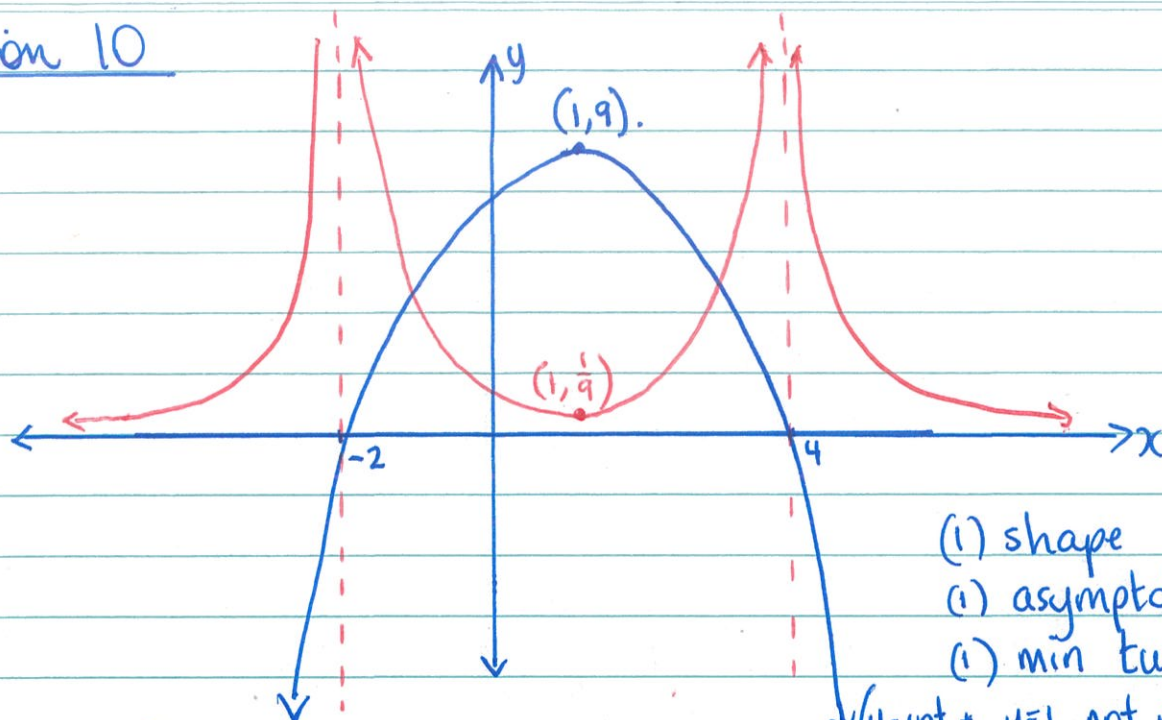
$3x = 6y - xy$

$3x = y(6-x)$

$y = \frac{3x}{6-x}$ (1)

Question 10

a)



- (1) shape
- (1) asymptotes.
- (1) min turning pt.
- (1) (y-int + y=1 not required)

b) * test true for $n=1$ $9^{1+2} - 4^1 = 729 - 4 = 725$ which is divisible by 5. (1)

$\therefore$ true for $n=1$

* Assume true for $n=k$

$$9^{k+2} - 4^k = 5M \text{ where } M \text{ is an integer} \quad (1)$$

* Prove true for $n=k+1$

$$\begin{aligned}
 9^{k+1+2} - 4^{k+1} &= 9^{k+3} - 4^{k+1} \\
 &= 9^{k+2} \cdot 9^1 - 4^{k+1} \\
 &= (5M + 4^k) \cdot 9 - 4^{k+1} \quad \text{from the assumption} \\
 &= 5 \cdot 9M + 9 \cdot 4^k - 4 \cdot 4^k \\
 &= 5 \cdot 9M + 4^k(9-4) \\
 &= 5 \cdot 9M + 4^k \cdot 5 \\
 &= 5(9M + 4^k) \text{ which is divisible by 5 (1)} \\
 &\quad \text{since } M \text{ \& } k \text{ are integers}
 \end{aligned}$$

$\therefore$ Statement proved true by Mathematical induction

(-1/2) missing 'from assumption' or equivalent

(-1/2) missing 'as $M \neq k$ are integers'

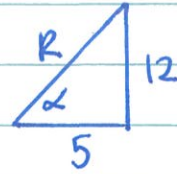
$$c) i) 5 \sin x + 12 \cos x = R \sin x \cos d + R \cos x \sin d.$$

$$R \cos d = 5$$

$$\cos d = \frac{5}{R}$$

$$R \sin d = 12$$

$$\sin d = \frac{12}{R}$$



(1) working towards

$$R^2 = 12^2 + 5^2$$

$$R = 13 \quad (1)$$

$$\tan d = \frac{12}{5}$$

$$d = 67.38013505$$

$$= 67^\circ \text{ (nearest degree). } (1)$$

$$5 \sin x + 12 \cos x = 13 \sin(x + 67^\circ).$$

$$ii) \text{ Maximum value is } 13. \quad (1)$$

$$iii) \quad 13 \sin(x + 67) = 13.$$

$$\sin(x + 67) = 1$$

$$x + 67 = 90^\circ$$

$$x = 23^\circ \quad (1)$$

$$d) \quad \underline{a} - \underline{b} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} - \begin{pmatrix} -m \\ -1 \end{pmatrix}$$

$$= (-3 + m)\underline{i} + 2\underline{j} \quad (1)$$

$$\underline{b}(\underline{a} - \underline{b}) = 0$$

$$-m(-3 + m) + (1 \cdot 2) = 0 \quad (1)$$

$$3m - m^2 - 2 = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m - 2)(m - 1) = 0$$

$$m = 1 \text{ or } 2. \quad (1)$$

Question 11

$$\begin{aligned} \text{a) i) } \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} \\ &= 2 \cos 2t \times \frac{1}{\sin 2t} \end{aligned}$$

$$\begin{aligned} y &= \sin 2t & x &= \sin^2 t \\ \frac{dy}{dt} &= 2 \cos 2t & \frac{dx}{dt} &= 2 \sin t \cos t \\ & & &= \sin 2t \end{aligned}$$

$$= \frac{2 \cos 2t}{\sin 2t} \quad (1).$$

$\sin t = \sqrt{x}$ $-\frac{1}{2}$ if no mention of why it is positive

$$= 2 \cot 2t$$

$\cos t = \sqrt{1-x}$ must mention domain as t why it's positive

$$\text{ii) } x = \sin^2 t \quad y = 2 \sin t \cos t$$

$$\cos t = \frac{y}{2 \sin t}$$

$$\cos^2 t = \frac{y^2}{4 \sin^2 t}$$

$$= \frac{y^2}{4x} \quad (1).$$

$$\sin^2 t + \cos^2 t = 1.$$

$$x + \frac{y^2}{4x} = 1. \quad (1).$$

$$4x^2 + y^2 = 4x.$$

$$y^2 = 4x - 4x^2 \quad \text{or} \quad 4x^2 - 4x + y^2 = 0. \quad (1).$$

$$b). \quad T c_k (x)^{7-k} \left(\frac{1}{ax^2}\right)^k.$$

$$x^{7-k} \times (x)^{-2k}$$

$$7-k + (-2k) = 1 \quad (1)$$

$$3k = 6$$

$$k = 2. \quad (1).$$

$$T c_2 (x)^5 \cdot \left(\frac{1}{ax^2}\right)^2 = \frac{3}{7} x$$

$$\frac{21 \cdot 1}{a^2} = \frac{3}{7}$$

(1).

$$a^2 = \frac{21 \times 7}{3}$$

$$a^2 = 49$$

$$a = \pm 7 \quad (1).$$

(2) for $a=7$.

$$c) i). \quad \text{proj}_{OA}^{OB} = \frac{9 \cdot 6 + 2 \cdot 8}{\sqrt{6^2 + 8^2}} (6\hat{i} + 8\hat{j}). \quad \left. \begin{array}{l} (\frac{1}{2}) \\ (1) \end{array} \right\}$$

$$= \frac{70}{100} (6\hat{i} + 8\hat{j}). \quad (\frac{1}{2}).$$

$$= \frac{21}{5} \hat{i} + \frac{28}{5} \hat{j}. \quad (-\frac{1}{2}) \text{ if left like this. } (\frac{1}{2})$$

$\therefore$ coordinates of C $(4.2, 5.6)$ or $(\frac{21}{5}, \frac{28}{5})$. ~~(1)~~ $(\frac{1}{2})$.

$$\text{ii). } |\vec{BC}| = \sqrt{(9-4 \cdot 2)^2 + (2-5 \cdot 6)^2}$$

$$= \sqrt{36}$$

$$= 6 \quad (1)$$

$$|\vec{OA}| = \sqrt{6^2 + 8^2}$$

$$= 10 \quad \left(\frac{1}{2}\right)$$

$$\text{Area } \triangle AOB = \frac{1}{2} \times 10 \times 6$$

$$= 30 \text{ u}^2 \quad \left(\frac{1}{2}\right).$$

$$\text{d). } \sum F \quad \text{horizontal} = -40 \cos \alpha + (-40 \cos \alpha) + 85$$

$$= 2 \times \left(-40 \times \frac{3}{4}\right) + 85$$

$$= -60 + 85$$

$$= +25 \quad (1).$$

$$\text{vertical} = 40 \sin \alpha + (-40 \sin \alpha) + 0$$

$$= 0 \quad (1)$$

$\therefore$ resultant vector has a magnitude of 25N to the right/east (1).

$\frac{1}{2}$ mag + $\frac{1}{2}$ direction
bearing T° required $\left(-\frac{1}{2}\right)$
○ not accepted. $\left(-\frac{1}{2}\right)$.

(-1) no vertical resolution
working out shown
or mention that they
cancel each other out.